

Tseegi Nyamdorj

Smith College
Northampton MA 01063
Phone: +1-413-275-3144

Web: <https://tseegi-n.github.io/>
e-mail: tnyamdorj@smith.edu

Education

Smith College, Northampton MA
B.A, Computer Science & Mathematical Statistics
GPA: 3.87/4.00, Dean's List 2021–2022

Sep 2021–May 2025

Awards / Honors

Middlebury College Project for Peace (\$10,000) - 2022
Smith College Universal Conference Fund (\$400) to attend CMSAC 2023
The Novice Champion in the 10th Asian Schools Debating Championship (ASDC) 2019
The Medal of Excellence issued by the Mongolian Government for outstanding study - 2019

Research / Industry Experience

AFS Mongolia

Jul 2020–Present

Summer Intern / Systems Manager / Board Member

- Optimized website SEO ranking by 207.3% using Google Search Console, Analytics, Tag Manager, HTML and CSS updates.
- Implemented GDPR compliance protocols as a certified Data Protection Officer.
- Developed and maintained website content for an international NGO organization.
- Led technical support for platforms: Salesforce, Trello, Workplace, and Google workspace.
- Managed a cohort of 10-15 summer interns on a weekly basis; conducted technical orientations.
- Conducted the first-ever SDGs summer camp in Ulaanbaatar, Mongolia through \$10,000 grant of Project for Peace (Middlebury College).

Smith College – Department of Computer Science

May 2024–Present

Undergraduate Research Assistant

- Conducted research in document engineering with Professors Halie Rando and Michael Robson.
- Developed a LLM pipeline using Llama3 for integrating track changes made in DOCX files into reproducible, collaborative manuscript writing workflows in Manubot (an open-source software project built on Markdown).
- Optimized model selection and fine-tuned prompt engineering for increased speed and accuracy with Promptfoo and Docker on HPC.
- Utilized Python, LLMs, and NLP techniques to automate document conversion.
- Presented research at the Computational Systems for Integrative Genomics (CSIG) 2024.

Carnegie Mellon University – Department of Statistics & Data Science

May 2023– August 2023

Undergraduate Researcher

- Conducted statistical research in sports analytics with Professor Ron Yurko.
- Developed a machine learning model using R to predict soccer match outcomes by analyzing betting and player performance metrics.
- Evaluated model performance with Brier score and calibration plots to improve prediction accuracy.
- Applied advanced techniques in data preprocessing, feature selection, and model evaluation to sports analytics.
- Presented results at the Carnegie Mellon Sports Analytics Conference (CMSAC) 2023.

Personal Projects**McDypar Conceptual Parsing Extension - CSC 390 Course Project**

Feb 2025–May 2025

- Extended the McDypar Lisp-based parser to handle recursive clause structures and disambiguation in natural language input.
- Added and tested new conceptual lexicons and sentence templates.
- Developed new demons (“expect,” “recursive,” etc.) to enable parsing of dependent clauses and ambiguous verbs.
- Integrated McDypar output with MARGIE’s BABEL paraphraser to generate natural language paraphrases of CD structures.

Baymax Arduino Companion - CSC 233 Course Project

Feb 2025–May 2025

- Designed and built an interactive Baymax robot using Arduino to respond to temperature and pain-level inputs.
- Integrated components such as LCD, DC motor, potentiometer, nMOS transistor, and pushbuttons to simulate Baymax’s “healthcare companion” behavior.
- Engineered physical movement using a motor-driven turntable and 3D-printed body parts.
- Developed IO mappings to display context-aware responses on an LCD based on user input.

Ghost Work Super Guide – CSC 356 Course Project

Feb 2025–Present

- Created a comprehensive 18-page guide for undergraduates navigating online crowd work, informed by survey data and HCI scholarship.
- Conducted qualitative research via surveys and online discourse analysis to understand student perceptions of ghost work.
- Performed platform analysis on AMT, Prolific, and Outlier AI, compiling onboarding tips, tools, plug-ins, and transparency tactics.
- Garnered 100+ unique readers and community feedback to refine the guide as a living resource.

CacheBite - CSC 256 Course Project

Feb 2024–May 2024

- Designed an app to connect students needing assistance (e.g., meal deliveries, mail pick-ups) with peers willing to help.
- Conducted data collection through surveys, observations; performed analysis using R.
- Developed user interfaces using XCode and Figma; delivered a functional demo on XCode.
- Addressed critical accessibility gaps, promoting inclusivity and connection on campus.

Minimax ChessBot on CUDA - CSC 220 / 290 Course Project

Sep 2024–Dec 2024

- Implemented a Minimax chessbot with alpha-beta pruning in C++ and CUDA.
- Utilized the SHL chess library to streamline board representation and move generation.
- Optimized computational efficiency by $5.7\times$ speed-up compared to Python-based implementations by parallelizing Minimax calculations.
- Delivered a functional chessbot capable of evaluating and making strategic moves efficiently.

Adults Pursuing Higher Education - SDS 291 Course Project

Sep 2024–Dec 2024

- Analyzed factors influencing adult education using logistic regression on IPUMS data
- Conducted data wrangling and visualization in R to explore relationships between age, parenthood, and educational enrollment.
- Developed and compared multiple statistical models to identify significant predictors, achieving robust hypothesis testing through chi-squared analysis.

Other Experience**Smith College – Department of Mathematical Sciences**

Sep 2022–Present

Teaching Assistant/ Course Grader

- Evaluated assignments and projects in Calculus I-III, Discrete Mathematics, and Mathematical Statistics seminar
- Provided feedback on student work and answered course-related questions

Food Rescue Network – Smith College

Oct 2021 – Dec 2023

Vice-President

- Spearheaded the founding and growth of the FRN. Led club activities and facilitated teamwork among board members.
- Analyzed food collection, weather patterns, and food surplus in Python, to develop food waste prevention reports for dining halls at the end of each semester.
- Reduced and helped redirect 15k pounds of food waste on campus in the first two years.
- Developed leadership skills through strategic planning, conflict resolution, and mentoring of other board members.

Joseph Metcalf High School – Mindich Fellowship

Jan 2022–Feb 2022

Teaching Assistant

- Selected to a cohort of Mindich fellows that engage in school-based teaching practica during the one-month apprenticeship
- Prepared social/ethnic studies and mathematics teaching materials for 6th graders and helped mentor teacher during classroom and recess